Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.25	20.15	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.02	2.01	-0.5%
power (W), mean	629	631	+0.2%
power (W), peak	652	672	+3.0%
energy (Wh)	212	212	-0.3%
energy, parked workers still on at idle power (Wh)	212	212	-0.3%
CPU used (cores), mean	0.900	0.920	+2.2%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.037	0.038	+2.2%
energy per core-hour (Wh)	700	686	-2.0%
node-hours per core-hour	6.67	6.52	-2.2%
pending pods, pod-minutes	0.267	0.000	-100.0%
pending pods, peak	1.00	0.000	-100.0%
HPA replicas, mean	8.99	9.09	+1.1%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.783	0.783	+0.0%
requests timed	1,035	1,145	+10.6%
time over the response line (% of samples)	1.26	0.000	-100.0%
response time (ms), mean	162	49.15	-69.7%
response time (ms), median	134	44.30	-66.9%
response time (ms), 95th percentile	316	72.38	-77.1%
response time (ms), 99th percentile	517	86.05	-83.4%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.2857	7.2803	-0.1%	[-1.9066, +1.8199]	EVEN, no significant difference
kwh_per_core_hour	0.7607	0.7607	-0.0%	[-0.1777, +0.1802]	EVEN, no significant difference
utilisation	0.0370	0.0380	+2.6%	[-0.0090, +0.0118]	EVEN, no significant difference
pending_pod_minutes_per_hour	0.7619	0.0000	-100.0%	[-2.2857, +0.0000]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	2.2442	2.3102	+2.9%	[-3.7142, +3.1444]	EVEN, no significant difference

What Omni-Compass did

• decisions: 20 (one per minute)

• HPA target writes: 0

• node-pool resizes: 3

• scheduling-floor adds (pending pods): 0

• power-cap pod resizes (in place): 0

• rollout actions: 0

• power cap decided per minute: 0.695

• heat (thermal) per minute: 0.358 0.392 0.42 0.445 0.467 0.486 0.502 0.518 0.531 0.542 0.552 0.558 0.563 0.567 0.572 0.576 0.579 0.583 0.584 0.586

• nodes decided per minute: 6 6 5 4 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3

• HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	615	617	5.00	5.00	0.000	0.000	0.524	0.547

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	619	618	5.00	5.75	0.000	0.000	0.613	0.581
2	6.00	6.00	619	620	5.75	6.00	0.000	0.000	0.606	0.632
3	6.00	6.00	624	626	6.00	6.00	0.250	0.000	0.742	0.789
4	6.00	6.00	635	636	9.00	9.50	0.000	0.000	1.04	1.05
5	6.00	6.00	636	639	9.67	10.00	0.000	0.000	1.08	1.14
6	6.00	6.00	636	632	10.00	10.00	0.000	0.000	1.06	0.937
7	6.00	6.00	647	656	10.00	10.00	0.000	0.000	1.37	1.43
8	6.00	6.00	645	654	10.00	10.00	0.000	0.000	1.31	1.59
9	6.00	6.00	646	657	10.00	10.00	0.000	0.000	1.37	1.63
10	6.00	6.00	641	647	10.00	10.00	0.000	0.000	1.34	1.37
11	6.00	6.00	619	615	10.00	10.00	0.000	0.000	0.621	0.520
12	6.00	6.00	619	616	10.00	10.00	0.000	0.000	0.599	0.541
13	6.00	6.00	624	622	10.00	10.00	0.000	0.000	0.755	0.720
14	6.00	6.00	637	635	10.00	10.00	0.000	0.000	1.09	1.07
15	6.00	6.00	633	631	10.00	10.00	0.000	0.000	0.988	0.945
16	6.00	6.00	636	633	10.00	10.00	0.000	0.000	1.03	1.03
17	6.00	6.00	626	624	10.00	10.00	0.000	0.000	0.801	0.740
18	6.00	6.00	620	616	10.00	10.00	0.000	0.000	0.626	0.533
19	6.00	6.00	619	619	10.00	10.00	0.000	0.000	0.638	0.617

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.